Insights & Updates on the Management of Multiple Sclerosis

Executive Summary:

Multiple sclerosis (MS) is a chronic inflammatory disease of the nervous system characterized by periodic or progressive immune-mediated demyelination.[1] With an estimated prevalence of 338 per 100,000 in the United States, MS is the most common neurological condition affecting adults worldwide without a cure.[2] Like other autoimmune disorders, MS manifests more frequently in women and mysteriously appears more often in populations residing at higher latitudes. Other environmental risk factors include obesity, tobacco exposure, and previous infection with mononucleosis.[1] Salient symptoms, like transverse myelitis, optic neuritis, and ataxia, resemble those found in other autoimmune conditions like lupus, sarcoidosis, and Sjogren's syndrome.[3] Consequently, misdiagnoses and delayed detection of MS remain problematic in clinical practice. In one study cohort, nearly one in five patients with an established MS diagnosis did not fulfill contemporary McDonald Criteria and most likely suffered from an alternative condition.[4] Although diagnostic delays have shrunk over the last few decades, four out of ten MS patients still report more than two years passing between symptom onset and diagnosis.[5]

Since MS remains a clinical diagnosis, health care providers may benefit from continuing medical education that enhances their abilities to detect the disease early. Mounting evidence indicates that early identification and intervention correlates with optimal outcomes for MS patients.[6,7,8] Traditional treatment options, like interferon-beta (IFN-β), can reduce the risk of disease progression by 33%.[9] Similarly, newer oral disease modifying therapies (DMTs) like fingolimod, teriflunomide, and dimethyl fumarate can slash annualized relapse rates by as much as 53% in relapsing-remitting MS.[10] As the number of available treatments multiply, neurologists and supporting staff should familiarize themselves with the most suitable treatment, which may vary in patients depending upon disease subtype, comorbid conditions, and pregnancy.[11] While newer generations of MS therapies eclipse their predecessors in some regards, their tolerability and method of administration can prove problematic for particular patient populations. By tailoring treatment plans according to patient preferences and circumstances, clinicians can secure better outcomes and adherence to disease modifying therapies.[7] Collectively, better MS education will help reduce the mounting direct medical costs and lost productivity associated with the disease, estimated to be in excess of $2.5 billion annually in the United States alone.[12]

Educational Objectives:

After participating in this activity, clinicians should be better able to:

- Recognize the signs and symptoms of MS to facilitate a prompter diagnosis and initiation of disease management therapies.
- Select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.
- Improve MS patient adherence to disease modifying therapies through an interdisciplinary approach to disease management.

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Educational Objective 1: Recognize the signs and symptoms of MS to facilitate a prompter diagnosis and initiation of disease management therapies.

Gap 1: Clinicians may struggle to recognize the signs and symptoms in MS patients necessary to make an early diagnosis, which is associated with improved outcomes.

Although validated diagnostic criteria exist for MS, difficulties identifying the disease remain common. This practice gap translates into misdiagnoses, delayed disease identification, and poorer patient outcomes. While the true incidence of MS misdiagnoses remains unknown, results from a pair of studies suggest that the rate may be as high as 1 in 5, with upwards of 4 in 10 symptomatic patients waiting two years or more for a diagnosis.[4,5] A separate study, which endeavored to better characterize the contemporary spectrum of MS misdiagnoses, observed that the problem is not confined to nonspecialists. Nearly a quarter (24%) of those enrolled in the study with a misdiagnosis received it from a neurologist with an MS-focused practice.[13] In this cohort, nearly a third of those examined carried a misdiagnosis for ten years or longer and two-thirds were exposed to immunomodulatory therapies. Beyond the unnecessary exposure to therapies with severe side effects, these patients endured substantial hardship financing costly drugs that failed to address their true, underlying condition. Participating MS specialists in this survey identified "clear evidence of an earlier missed opportunity to make a correct diagnosis" in a majority (72%) of patients.[13]

Accumulating evidence also indicates that early intervention most effectively preserves neurological function by preventing future deterioration.[14] Time to the expanded disability status scale (EDSS) was significantly longer in patients receiving early treatment with interferon-beta (IFN-β) versus placebo in the 2-year blinded phase of the BENEFIT study.[15] Study authors concluded that the risk of progression was reduced by 40% and that a single clinical event could influence the accumulation of further disability. These results corroborate conclusions drawn in other IFN-β studies that together showed how early intervention reduces the probability of patients exhibiting an event progressing to clinically definite MS.[16] Prompt treatment with glatiramer acetate (GA), teriflunomide, and alemtuzumab also confers protection from disease progression.[8] In an extension phase of the FREEDOMS study, patients in the placebo arm switched to fingolimod experienced a reduction in relapse rates, disability progression, and brain atrophy. That said, gains made by patients in the fingolimod arm dwarfed those made by patients later switched from placebo to the new therapy.[8]

The differential diagnosis of MS often remains one of exclusion, and its complexity can delay clinician efforts to identify the disease. In practice, MS manifests as one of four subtypes that differ on the basis of progression: relapsing remitting, secondary progressive, primary progressive, and progressive relapsing.[7] For all forms, accumulating sensorimotor and cognitive deficits trace their origins to immune-mediated demyelination of neurons in the central nervous system (CNS).[17] Accumulating axonal damage in the CNS results in a diverse array of symptoms, many of which overlap with those found in other neurological disorders. Common presenting syndromes, like optic neuritis, limb weakness, and ataxia, can manifest in other diseases like neuromyelitis optica, vasculitis, and stroke; fatigue, bladder dysfunction, and cognitive decline can also occur in other maladies like fibromyalgia, Lyme disease, and dementia.[3] MS' resemblance to a broad range of disorders, coupled with a notoriously variable disease course, can make the neurodegenerative condition tough to quickly identify, particularly in patients with preexisting conditions.[18]

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White matter demyelination, a hallmark of early disease, occurs heterogeneously and appears as lesions when visualized through the use of magnetic resonance imaging (MRI).[1] To make a diagnosis of MS, clinicians refer to the McDonald criteria, which requires evidence of disseminated damage over time and space (Table 1). Conventional MRI techniques, like T2-weighted imaging and gadolinium-enhanced T1-weighted imaging constitute the gold standard for disease identification and monitoring, as they have high sensitivity in assessing the evolution of demyelinating plaques. Beyond neuroimaging, clinicians should also order blood work and analyze the content of cerebrospinal fluid (CSF); the presence of oligoclonal bands is a characteristic finding present in upwards of 90% of MS patients.[1] That said, these antibodies are not pathognomonic for disease. Another useful molecular biomarker, albeit one with lower sensitivity, is the immunoglobulin (Ig) G index, which is increased in about 70% of MS patients.[1] An IgG index value greater than 0.7 indicates increased intrathecal B cell response and disease activity in the CNS. Further biomarkers for MS, particularly those with prognostic value, remain lacking.[10]

Table 1: The McDonald Criteria for identifying multiple sclerosis (MS) in clinical practice.

Clinical Presentation	Additional Criteria to Make MS Diagnosis
• 2 or more attacks and clinical evidence of 2 or more lesions; • 2 or more attacks and clinical evidence of 1 lesion with historical evidence of prior attack involving lesion in different location	None. DIS and DIT have been met.
• 2 or more attacks and clinical evidence of 1 lesion	DIS shown by one of these criteria: • Additional clinical attack implicating different CNS site. • 1 or more MS-typical T2 lesions in 2 or more CNS areas: periventricular, cortical, juxtacortical, infratentorial, or spinal cord.
• 1 attack and clinical evidence of 2 or more lesions	DIT shown by one of these criteria: • Additional clinical attack • Simultaneous presence of both enhancing and non-enhancing MS-typical MRI lesions, or new T2 or enhancing MRI lesion compared to baseline scan (without regard to timing of baseline scan) • CSF oligoclonal bands
• 1 attack and clinical evidence of 1 lesion	DIS shown by one these criteria: • Additional attack implicating different CNS site • 1 or more MS-typical T2 lesions in 2 or more areas of CNS: periventricular, cortical, juxtacortical, infratentorial, or spinal cord **AND** DIT shown by one of these criteria: • Additional clinical attack • Simultaneous presence of both enhancing and non-enhancing MS-typical lesions, or new T2 or enhancing MRI lesion compared to baseline scan (without regard to timing of baseline scan) • CSF oligoclonal bands

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DIT = dissemination in time, CNS = central nervous system, CSF = cerebrospinal fluid,
DIS = dissemination in space, T2 lesion = hyperintense lesion on T2-weighted MRI

Continuing medical education can strengthen clinician familiarity with symptoms suggesting or refuting a diagnosis of MS. Since MS is most often confused for migraine, fibromyalgia, nonspecific or nonlocalizing neurologic symptoms with abnormal MRI, and neuromyelitis optica, educational initiatives should logically emphasize the characteristics distinguishing one condition from another.[13] In particular, expanded neuroimaging and comparative MRI criteria could help clinicians readily differentiate between other conditions of the CNS.[18] To expedite the time to diagnosis, neurologists and supporting staff alike would also benefit from greater knowledge of "red flags" that immediately suggest an alternative diagnosis.[19] Red flags suggestive of alternative conditions include bone lesions, retinopathy, polyradiculopathy, hematologic manifestations, lung or renal involvement, and lesion enlargement.[20] Finally, better adherence to established diagnostic criteria and avoiding overreliance on MRI imaging abnormalities can improve MS detection rates.[13] Although neuroimaging forms the core of any successful diagnosis, clinicians should remain mindful of other important signs like cognitive decline and depression that manifest commonly in MS patients.[21]

Educational Objective #2: Select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.

Gap #2: Clinicians may not be able to select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.

As the number of available therapies for MS multiplies, so too does the need for clinicians to understand which treatments are best in certain contexts.[14] Although all FDA approved disease modifying therapies (DMTs) can reduce relapses and disease progression, each class has unique side effect profiles. When evaluating the suitability of a treatment regimen, clinicians should consider disease severity and phenotype, along with patient preferences, risk tolerance, and existing comorbidities.[10] Revised treatment guidelines from the American Academy of Neurology (AAN) also recommend different drugs depending upon disease subtype. While options abound to manage relapsing-remitting MS, only ocrelizumab has FDA approval to treat primary progressive MS.[14] Finally, since young women constitute a majority of MS patients, health care providers may also benefit by knowing which DMTs are safest for use during pregnancy and breastfeeding.[11]

First approved in 1993, interferon beta (IFN-β) therapies constituted the first class of treatments for MS patients and remain relevant today.[22] IFN-β therapies have several known mechanisms of action, including downregulation of the major histocompatibility complex (MHC) class II expressed on antigen-presenting cells, induction of IL-10 production by T-cells, and inhibition of T-cell migration. These changes are mediated by transcription factors that migrate to the nucleus and modulate the expression of over 500 genes that encode elements of the human immune response.[22] As the oldest category of treatment, IFN-β therapies have been extensively validated through several clinical and comparative trials. IFN-β-1a administered thrice weekly at a dose of 44 μg and IFN-β-1b taken every other day at 250 μg have been found to reduce relapses by a third compared with a placebo.[10] Although newer oral DMTs outperform IFN-β therapies, these older immunomodulators are well-tolerated, with side effects limited to injection-site reactions, flu-like symptoms, and a heightened risk of depression.[10,22]

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Oral DMTs like fingolimod, teriflunomide, and diroximel fumarate can all improve annualized relapse rates (ARRs) but generally carry a higher risk of severe side effects. Fingolimod, which first gained FDA approval in 2010, selectively binds sphingosine-1-phosphate (S1P1) receptors expressed by neurons, astrocytes, and oligodendrocytes.[10] As a functional antagonist, fingolimod blocks lymphocytic egress and exerts neuroprotective effects by preventing inflammatory cytokines like TNF-α, IL-6, and IL-17 from circulating in the CNS. Since the drug decreases systemic lymphocyte counts, patients on fingolimod have a higher risk of opportunistic infections (e.g. herpes). Other safety concerns, like bradycardia, macular edema, and hypertension, result from the ubiquity of S1P1 receptors throughout the circulatory system.[10] Recommendations for the treatment, administered once daily, vary by region; it is restricted as a first-line therapy in many European countries but available at any disease stage in the United States.

Teriflunomide, the second MS oral agent to gain FDA approval, lowers lymphocyte production by inhibiting pyrimidine synthesis in dividing cells. In the TEMSO and TOWER phase III clinical trials, participants in the high dose teriflunomide treatment arms had reduced annualized relapse rates compared to placebo and a lower risk of disability progression after 12 weeks.[10] Results from the TENERE study suggest that the 14 mg. dosage of teriflunomide has comparable efficacy to high-dose IFN and glatiramer acetate (GA).[10] Side effects include abnormal liver enzyme levels, hair thinning, diarrhea, nausea, tuberculosis reactivation, and teratogenicity. Considering these issues, teriflunomide is contraindicated in pregnant patients, those with hepatic injury, and those with a positive screen for tuberculosis.[10,11]

Dimethyl fumarate exerts neuroprotective effects in MS patients by activating nuclear factor erythroid 2 related factor, which ultimately results in decreased inflammation. In the randomized, double-blind DEFINE trial, patients taking dimethyl fumarate twice daily at a dose of 240 mg. had a reduction in the number of new T2 or gadolinium-enhanced lesions. Compared with those in the placebo arm of the trial, patients taking dimethyl fumarate also had a reduced annualized relapse rate (53%) and progression of disability (38%).[10] During these investigations, patient reported tolerability issues included gastrointestinal discomfort (20-25%) and flushing (25-30%); other risks of the drug include severe lymphopenia and progressive multifocal leukoencephalopathy.[10,23] Although discomfort typically subsides after one month of treatment, clinicians may want to exercise caution prescribing dimethyl fumarate in patients with baseline GI issues, as the drug could exacerbate symptoms caused by other diseases (e.g. irritable bowel syndrome).[10]

Since MS typically afflicts younger women, clinicians should be familiar with DMTs suitable for use during pregnancy and breastfeeding.[11] Current treatment guidelines from the AAN generally advise against the use of DMTs during pregnancy because their safety during the second and third trimesters remains unknown. Pregnancy typically lowers the risk of relapse in MS patients, although treatment may be warranted in patients still experiencing severe attacks.[14] Of the available options, only glatiramer acetate (GA) carries a pregnancy category rating of B. Alternatives have pregnancy category ratings of C, and teriflunomide has an X rating because of its known teratogenicity.[10] Additionally, GA does not require regular laboratory monitoring and observational studies have noted that it does not appear to induce congenital defects, miscarriages, or prematurity. The rapid degradation of GA following its administration also suggests that it would not linger in breast milk.[24] Consequently, some physicians see no harm continuing GA for patients during and after pregnancy.

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Educational Objective #3: Improve MS patient adherence to disease modifying therapies through an interdisciplinary approach to disease management.

Gap #3: Clinicians need assistance identifying MS patients at risk of skipping medication and providing the resources necessary to prevent treatment non-adherence.

Although a panoply of disease-modifying therapies (DMTs) have gained FDA approval to manage MS in the last decade, these therapeutic strategies are only effective if taken as prescribed.[25] Studies assessing real-world DMT-usage patterns collectively suggest suboptimal adherence to both injectable and oral-administered therapies. According to one economic study, approximately 40% of enrolled patients (n=12,431) failed to meet adherence criteria (PDC > 0.8) when examined over a period of 1-5 years after commencing treatment.[26] This estimate corroborates findings from a separate analysis that concluded that 42% of MS patients taking oral DMTs were non-adherent; 45% of those taking self-injectables fell into the same category.[27] A more optimistic projection, relying on alternative methodologies, still pegged the non-adherence rate at 23%.[28]

Widespread patient failure to commit to treatment regimens has dire economic and personal consequences. Overwhelming evidence from the scientific literature indicates that a robust relationship exists between adherence and a lower likelihood of severe relapse in MS ($P < 0.0001$).[26,29] Specifically, in one study cohort, adherence to DMT reduced the likelihood of relapse by 42%, hospitalizations by 52%, and emergency visits by 38%.[26] These continued interventions to treat relapses sap patient productivity, raise healthcare costs, and divert limited resources away from other patient populations requiring care. Using the differences in predicted mean costs, these study authors also concluded that adherence would decrease total annual medical care costs by $5,816 per patient.[26] Taken together, the cost burden of MS is upwards of $10 billion in the United States.[12]

To improve outcomes and alleviate increasingly overstretched healthcare systems, clinicians should strive to improve MS patient adherence to DMTs.[7,26] Since adherence is a multifactorial problem, health care providers must become familiar with the salient barriers that predict patient delinquency. Most research in this area agrees that sex, age, cognitive capacity, and comorbidities all influence patient likelihood to follow prescriptions.[25,30] More specifically, investigators have observed that younger patients under 45, women, and those suffering from greater cognitive impairment are all more likely to be non-adherent.[7] A majority of patients in a study on DMT adherence most commonly cited forgetfulness as a primary factor behind missed doses.[31] Major depressive disorder, among the most common comorbid conditions in MS, is also another significant predictor of nonadherence.[7]

Less concordance exists around the importance of treatment type, with conflicting reports of nonadherence rates for oral and injectable DMTs.[25] Oral agents, like fingolimod, teriflunomide, and diroxmel fumarate, tend to have better usage rates over their injectable counterparts, as the latter modality can cause uncomfortable reactions at the insertion site.[7] That said, patient preferences have been noted to shift towards injectable alternatives when oral DMTs must be administered more frequently (thrice weekly) and result in more severe side effects.[32] As such, clinicians should consider factors like convenience and patient preferences when determining suitable treatment options.[33]

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Going forward, an interdisciplinary approach to MS care may prove key in improving DMT adherence rates.[34] Using the factors noted above, clinicians can help guarantee that patients receive support and additional education commensurate to their risk of delinquency. More subjective variables, like patient-physician rapport and self-efficacy, are also known to improve outcomes as patients feel more capable of coping with their condition.[35] During initial meetings and follow-ups, neurologists and supporting staff should promote patient efforts to understand MS and set expectations for treatments. While work continues to develop newer, more effective drugs, there remains no cure for MS; existing DMTs act to manage symptoms and abate further neurological deterioration.[1] Between physical meetings, nurses can also supply telephone-based counseling and remote support to encourage patients to continue with treatment. Several studies have demonstrated the value of this sustained support for MS patients and for others battling cardiovascular disease and diabetes.[34]

Other practical matters, like treatment costs and emotional support, are best handled by a blend of social workers, therapists, and family members.[36] Social workers can serve as generalists that educate, console, and advocate on behalf of MS patients who may lack the initiative to seek further treatment and financial resources for continued DMTs. In particular, these professionals can help MS patients navigate labyrinthine insurance plans and select the most affordable option; higher costs out-of-pocket to patients unsurprisingly correlate with higher treatment nonadherence.[37] Therapists, in contrast, can specifically address lingering mental health conditions like anxiety and depression that can derail efforts to observe physician orders. Finally, family members who actively support loved ones battling disease can fill any remaining cracks in education or advocacy. Their sustained support may help MS patients taper dependence on paid, external support systems and reintegrate successfully into the wider world.

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Gaps, Objectives, and Expected Outcomes

Educational Gap	Educational Objective	Practice Goal	Outcome Levels
Clinicians may struggle to recognize the signs and symptoms in MS patients necessary to make an early diagnosis, which is associated with improved outcomes.	Recognize the signs and symptoms of MS to facilitate a prompter, accurate diagnosis and initiation of disease management therapies.	Learners will have greater competence to recognize the signs and symptoms of MS to facilitate a prompter diagnosis and initiate disease management therapies.	Level 1: Participation Level 2: Satisfaction Level 3: Declarative Knowledge Level 4: Competence
Clinicians may not be able to select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.	Select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.	Learners will have greater competence to select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.	Level 1: Participation Level 2: Satisfaction Level 3: Declarative Knowledge Level 4: Competence
Clinicians need assistance identifying MS patients at risk of skipping medication and providing the resources necessary to prevent treatment non-adherence.	Improve MS patient adherence to disease modifying therapies through an interdisciplinary approach to disease management.	Learners will have greater competence to improve MS patient adherence to disease modifying therapies by using an interdisciplinary approach to disease management.	Level 1: Participation Level 2: Satisfaction Level 3: Declarative Knowledge Level 4: Competence

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Agenda

Topic	Key Purpose	Duration
Welcome, program goals, and pre-assessment		5 min.
Overview of signs and symptoms suggestive of a multiple sclerosis diagnosis • Common symptoms of MS • MS mimics and potential misdiagnoses – neuromyelitis optica, fibromyalgia, migraine • MRI evidence & McDonald criteria for establishing and confirming MS diagnoses.	Recognize the signs and symptoms of MS to facilitate a prompter diagnosis and initiation of disease management therapies.	15 min.
Overview of treatment strategies for multiple sclerosis: • Injectable DMTs – interferon-β • Oral DMTs – fingolimod, teriflunomide, diroximel fumarate • Options during pregnancy – glatiramer acetate	Select the optimal treatment strategy for MS patients on the basis of status and demonstrated disease progression.	15 min.
Overview of non-adherence risk factors and roles of various professionals in MS management: • Age, sex, cognitive state, comorbid conditions, treatment types. • Patient-physician communication, education, promoting self-efficacy. • Nurses as remote counselors. • Non-medical professionals and external support systems.	Improve MS patient adherence to disease modifying therapies through an interdisciplinary approach to disease management.	15 min.
Post-test Q&A	Clarifying questions	10 min.

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